

ASSEMBLY OF THE LOCAL TYPE II EXCLUSION THEOREM FOR NAVIER-STOKES

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ABSTRACT. We assemble the local Type II exclusion argument for the three-dimensional Navier–Stokes equations from the local results proved in the preceding papers. The entry point is the local Caffarelli–Kohn–Nirenberg concentration theorem: any singular point which is not in the local Type I alternative yields a positive local Type II concentration sequence. The repaired-gauge construction places each retained nondegenerate core into a renormalized local Navier–Stokes chart. The single-core, multibubble/cascade, rough-core, scale-collapse, and virial-controlled terminal analyses then exhaust or eliminate the local state-space alternatives to which they apply. The proof below uses only those local alternatives and does not add any stationary-profile classification theorem.

1. LOCAL ENTRY

Let (u, p) be a suitable weak solution of the three-dimensional Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

on $\mathbb{R}^3 \times (T - \delta, T)$. For $z_0 = (x_0, T)$ and $r > 0$, set

$$Q_r(z_0) = B_r(x_0) \times (T - r^2, T),$$

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt,$$

and

$$D(z_0, r) = r^{-2} \int_{T-r^2}^T \int_{B_r(x_0)} |p(x, t) - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

The pressure is always taken modulo functions of time, and the spatial mean normalization is the one used in the Caffarelli–Kohn–Nirenberg criterion.

Definition 1.1 (Singular set and local Type II point). Let $\Sigma(T)$ be the set of points x_0 for which u is not locally bounded in any backward cylinder $Q_r(x_0, T)$. A point $x_0 \in \Sigma(T)$ is in the local Type II alternative if it carries a positive local Caffarelli–Kohn–Nirenberg concentration sequence and no local Type I scale-invariant bound holds at (x_0, T) .

Definition 1.2 (Positive local Type II concentration sequence). A sequence $r_n \downarrow 0$ is a positive local Type II concentration sequence at (x_0, T) if there exists $\eta > 0$ such that

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n,$$

and (x_0, T) is in the local Type II alternative. The associated parabolic rescalings are

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \quad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Theorem 1.3 (Local concentration and Type I/Type II dichotomy). *If $\Sigma(T) \neq \emptyset$, then there are $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

At the concentration point exactly one of the following alternatives holds: there is a local Type I scale-invariant bound, or (x_0, T) is in the local Type II alternative.

Proof. This is the local concentration and blow-up dichotomy proved in Paper 0. $\square$

2. LOCAL THEOREMS USED IN THE ASSEMBLY

We record the local results from the preceding Type II papers in the form used below. They are quoted as PDE statements and are not reproved in this paper.

Theorem 2.1 (Repaired-gauge representation). *Let a positive local Type II concentration sequence contain a nondegenerate selected core. Then, after passing to selected windows, there are local scale-center variables $\lambda(\tau) > 0$, $x_c(\tau)$, renormalized fields*

$$V(y, \tau) = \lambda(\tau)u(x_c(\tau) + \lambda(\tau)y, t(\tau)), \quad P(y, \tau) = \lambda(\tau)^2 p(x_c(\tau) + \lambda(\tau)y, t(\tau)) + \pi(\tau),$$

and modulation coefficients a, b such that

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0.$$

On each compact renormalized window one has the local suitable energy inequality, local pressure reconstruction modulo functions of time, stability under subsequences, and the compact-window bounds proved in the repaired-gauge paper. Failure of the nondegenerate scale-center selection is part of the multibubble, cascade, or scale-rigidity alternatives.

Proof. This is the compact-window representation package proved in Paper 2. $\square$

Theorem 2.2 (Compact single-core exclusion). *No compact single-core vanishing-dissipation branch can be a local Type II sequence. More explicitly, let a represented selected-window sequence have:*

- (i) *positive local CKN concentration on a fixed compact cylinder;*
- (ii) *a uniformly bounded compact-cylinder suitable package $A + E + C + D$;*
- (iii) *nondegenerate local scale-center representation;*
- (iv) *bounded modulation coefficients;*
- (v) *local pressure reconstruction and pressure stability modulo spatial means;*
- (vi) *vanishing localized dissipation on some retained compact cylinder.*

Then the sequence is incompatible with the local Type II condition.

Proof. This is the local single-core Type II criterion proved in Paper 1. $\square$

Theorem 2.3 (Multibubble and cascade exclusion). *Assume the terminal admissibility and scale-rigidity hypotheses stated in the multibubble/cascade analysis. Then no local multibubble, cascade, or gauge-degenerate Type II branch can occur in the class produced by the local active-core extraction. Equivalently, every failure of the single nondegenerate compact-core alternative is excluded by the local multibubble/cascade analysis under those stated hypotheses.*

Proof. This is the local multibubble and cascade exclusion theorem, together with its removal corollary, proved in Paper 3. $\square$

Theorem 2.4 (Rough-core reduction). *On any compact renormalized window satisfying the local energy inequality and bounded modulation hypotheses, finite outer velocity-pressure CKN control $\mathcal{C} + \mathcal{D}$ implies finite inner windowed H^1 control. Hence failure of local windowed H^1 control on a retained inner cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ package on every relevant enclosing cylinder.*

Proof. This is the rough-core exclusion on compact windows proved in Paper 4. $\square$

Theorem 2.5 (Finite-cost scale-collapse exclusion). *No represented branch with finite scale-collapsing cost, and hence no branch with finite absolute scale cost, can have genuine scale collapse together with a nonvanishing selected core.*

Proof. This is the scale-collapse cost exclusion and absolute scale-cost exclusion of Paper 5. $\square$

Theorem 2.6 (Scale-collapse state-space stratification). *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. After passing to selected terminal windows, exactly one of the state-space alternatives isolated in Paper 5 occurs:*

- (i) *finite scale-collapsing cost;*
- (ii) *finite absolute scale cost;*
- (iii) *thin-drift defect;*
- (iv) *autonomous-modulation defect;*
- (v) *compactness defect;*
- (vi) *a nonzero autonomous reduced limit in the scale-rigid state.*

In the local Type II state space used by the multibubble/cascade and scale-collapse analyses, alternatives (iii)–(vi) are not new compact single-core mechanisms: the thin-drift, modulation, and compactness defects are exits from the compact finite-cost scale-collapse branch, and the scale-rigid state is the state handled by the scale-rigidity hypothesis in the multibubble/cascade theorem.

Proof. This is the scale-collapse state-space stratification, combined with the scale-rigidity branch used in the multibubble/cascade reduction. $\square$

Definition 2.7 (Virial-dissipative terminal state). Let $G_\nu(y) = e^{-|y|^2/(4\nu)}$. A scale-rigid terminal state is called velocity-virial dissipative if its selected terminal representative (V, P) satisfies, after the usual pressure normalization, the autonomous self-similar equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \nabla \cdot V = 0,$$

and the Gaussian identity below is justified by cutoff approximation on the selected renormalized windows. It is called swirl-virial dissipative if, in addition, the representative is axisymmetric, $\Gamma = RV_\theta$ is the scale-invariant circulation, and the corresponding weighted circulation identity below is justified.

Theorem 2.8 (Virial exclusion for dissipative terminal states). *Let a scale-rigid terminal state satisfy the hypotheses of Theorem 2.7 and be bounded for all negative renormalized times in the sense needed for the Gaussian integrals below.*

- (a) *Suppose the velocity virial defect is absorbable: for some $\kappa > 0$ and a.e. τ ,*

$$-\frac{1}{2\nu} \int_{\mathbb{R}^3} |V|^2 (V \cdot y) G_\nu dy - \frac{1}{\nu} \int_{\mathbb{R}^3} P (V \cdot y) G_\nu dy \leq (1 - \kappa) \int_{\mathbb{R}^3} |V|^2 G_\nu dy + 2\nu(1 - \kappa) \int_{\mathbb{R}^3} |\nabla V|^2 G_\nu dy.$$

Then

$$\int_{\mathbb{R}^3} |V(y, \tau)|^2 G_\nu(y) dy = 0 \quad \text{for every } \tau.$$

Consequently no retained terminal core with positive Gaussian velocity mass can belong to this branch.

- (b) *Suppose the terminal state is axisymmetric and the circulation defect is absorbable: for some $\kappa > 0$ and a.e. τ ,*

$$-\frac{1}{2\nu} \int_{\mathbb{R}^3} \Gamma^2 (V \cdot y) G_\nu dy \leq (1 - \kappa) \int_{\mathbb{R}^3} \Gamma^2 G_\nu dy + 2\nu(1 - \kappa) \int_{\mathbb{R}^3} |\nabla \Gamma|^2 G_\nu dy.$$

Then

$$\int_{\mathbb{R}^3} \Gamma(y, \tau)^2 G_\nu(y) dy = 0 \quad \text{for every } \tau.$$

Thus every retained axisymmetric terminal core whose remaining obstruction is a positive Gaussian circulation mass is excluded by the weighted circulation identity.

Proof. For the velocity statement set

$$\mathcal{E}(\tau) = \int_{\mathbb{R}^3} |V(y, \tau)|^2 G_\nu(y) dy.$$

The Gaussian virial identity gives

$$\partial_\tau \mathcal{E} = -2\nu \int_{\mathbb{R}^3} |\nabla V|^2 G_\nu dy - \int_{\mathbb{R}^3} |V|^2 G_\nu dy - \frac{1}{2\nu} \int_{\mathbb{R}^3} |V|^2 (V \cdot y) G_\nu dy - \frac{1}{\nu} \int_{\mathbb{R}^3} P(V \cdot y) G_\nu dy.$$

The absorbability hypothesis therefore implies

$$\partial_\tau \mathcal{E} + \kappa \mathcal{E} + 2\nu \kappa \int_{\mathbb{R}^3} |\nabla V|^2 G_\nu dy \leq 0.$$

In particular $\partial_\tau \mathcal{E} + \kappa \mathcal{E} \leq 0$. Integrating from $s < \tau$ to τ gives

$$\mathcal{E}(\tau) \leq e^{-\kappa(\tau-s)} \mathcal{E}(s).$$

The ancient boundedness of the terminal state makes $\sup_{s < \tau} \mathcal{E}(s) < \infty$. Sending $s \rightarrow -\infty$ yields $\mathcal{E}(\tau) = 0$.

For the axisymmetric statement set

$$\mathcal{G}(\tau) = \int_{\mathbb{R}^3} \Gamma(y, \tau)^2 G_\nu(y) dy.$$

The weighted circulation identity is

$$\partial_\tau \mathcal{G} + \mathcal{G} + 2\nu \int_{\mathbb{R}^3} |\nabla \Gamma|^2 G_\nu dy = -\frac{1}{2\nu} \int_{\mathbb{R}^3} \Gamma^2 (V \cdot y) G_\nu dy.$$

The assumed absorption gives

$$\partial_\tau \mathcal{G} + \kappa \mathcal{G} + 2\nu \kappa \int_{\mathbb{R}^3} |\nabla \Gamma|^2 G_\nu dy \leq 0.$$

The same backward Gronwall argument gives $\mathcal{G}(\tau) = 0$ for every τ . $\square$

Remark 2.9 (Controlled swirl alone). If the axisymmetric representative satisfies $\Gamma \in L^\infty$, the same weighted circulation identity gives the a priori estimate

$$\mathcal{G}(\tau)^{1/2} \lesssim \|V\|_{L^\infty} \|\Gamma\|_{L^\infty}$$

for bounded ancient terminal states. This estimate is useful in the local state-space stratification, but by itself it is not an exclusion. The exclusion above requires the stated absorption, vanishing defect, or an equivalent condition which makes the virial inequality dissipative.

3. STATE-SPACE ASSEMBLY

Theorem 3.1 (Local Type II state-space decomposition). *Every positive local Type II concentration sequence in [Theorem 1.2](#), after passing to subsequences and selected windows in the local state-space analysis, enters one of the following alternatives:*

- (i) *a compact single-core vanishing-dissipation branch;*
- (ii) *a multibubble, cascade, or gauge-degenerate branch;*
- (iii) *a rough-core branch, meaning loss of local windowed H^1 control on a retained compact cylinder;*

- (iv) a finite-cost scale-collapse branch with nonvanishing selected core;
- (v) a scale-rigid state covered by the scale-rigidity hypothesis in [Theorem 2.3](#).

In the last alternative, any terminal subbranch satisfying [Theorem 2.8](#) is eliminated before the remaining scale-rigid state is passed to the scale-rigidity hypothesis.

Proof. This is the local state-space decomposition obtained by combining [Theorems 2.1, 2.3, 2.4](#) and [2.6](#). The representation theorem gives the nondegenerate single-core chart whenever a unique retained core is selected. Failure of that selection, splitting of positive CKN mass, compactness failure, or scale cascade is precisely the multibubble/cascade side. On a represented retained core, rough-core loss is equivalent by [Theorem 2.4](#) to failure of compact CKN control on an enclosing cylinder, hence returns to the multibubble/cascade side. Genuine scale collapse is divided by [Theorem 2.6](#) into finite-cost scale collapse or one of the scale-rigid terminal states used in [Theorem 2.3](#). These cases exhaust the local alternatives. $\square$

Proposition 3.2 (Elimination of local Type II states). *No positive local Type II concentration sequence can lie in any of the alternatives (i)–(v) of [Theorem 3.1](#).*

Proof. Alternative (i) is excluded by the compact single-core criterion, [Theorem 2.2](#).

Alternative (ii) is excluded by the local multibubble and cascade theorem, [Theorem 2.3](#), under its terminal admissibility and scale-rigidity hypotheses.

Alternative (iii) is not an independent terminal state. By [Theorem 2.4](#), rough-core loss of local H^1 control forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ package on enclosing compact cylinders. Such failure produces an active compact-cylinder concentration branch and therefore belongs to the multibubble/cascade alternative, already excluded by [Theorem 2.3](#).

Alternative (iv) is excluded by the finite-cost scale-collapse theorem, [Theorem 2.5](#).

Alternative (v) is the scale-rigid terminal state in [Theorem 2.3](#). If this terminal state is velocity-virial dissipative, [Theorem 2.8](#) forces its Gaussian velocity mass to vanish, which contradicts retention of a positive terminal core. If it is axisymmetric and swirl-virial dissipative with positive retained circulation mass, the same theorem forces $\Gamma \equiv 0$, contradicting that positive circulation mass. The remaining scale-rigid terminal states are precisely those left to the scale-rigidity hypothesis in [Theorem 2.3](#). Under that hypothesis, the state either has a nonzero bounded selected-window limit, which is incompatible with the local Type II condition, or it reduces to the compact single-core branch, which is excluded by [Theorem 2.2](#). Hence it cannot be a local Type II branch. $\square$

4. ASSEMBLY THEOREM

Theorem 4.1 (Local Type II exclusion). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. Then no point of $\Sigma(T)$ lies in the local Type II alternative, within the local state-space class covered by [Theorem 3.1](#).*

Proof. Assume, for contradiction, that $x_0 \in \Sigma(T)$ lies in the local Type II alternative. By [Theorem 1.3](#), there is a positive local Type II concentration sequence $r_n \downarrow 0$ at (x_0, T) . The local state-space decomposition, [Theorem 3.1](#), places this sequence, after passing to selected windows, in one of the five alternatives listed there. Each alternative is impossible by [Theorem 3.2](#). This contradiction proves that no such x_0 exists in the covered local state-space class. $\square$

Corollary 4.2 (Final local dichotomy after Type I separation). *Every singular point at time T in the covered local state-space class belongs to the local Type I alternative. Equivalently, after the Type I case is handled by the Type I part of the theory, the local Type II state space contains no remaining singular branch in that class.*

Proof. Let $x_0 \in \Sigma(T)$. By [Theorem 1.3](#), a concentration point is either in the local Type I alternative or in the local Type II alternative. [Theorem 4.1](#) excludes the latter. $\square$

5. CONCLUSION

The proof uses only local PDE statements. The local concentration theorem supplies positive local CKN concentration and the Type I/Type II dichotomy. The repaired-gauge theorem supplies the local renormalized representation for nondegenerate selected cores. The single-core theorem excludes compact single-core vanishing-dissipation branches. The multibubble/cascade theorem removes splitting, cascades, gauge degeneracy, and the scale-rigid terminal state under its stated local hypotheses. The virial identities eliminate the velocity-virial dissipative terminal states and the axisymmetric swirl-virial dissipative terminal states before the remaining scale-rigid alternatives are handled by that hypothesis. The rough-core estimate sends loss of windowed H^1 control back to failure of compact CKN control. The scale-collapse theorem excludes finite-cost scale collapse and stratifies the remaining scale-collapse states locally. Thus the local state-space alternatives are exhausted without adding any further classification theorem.